

French→English Transformer trained from scratch: where it generalizes and where it breaks

GitHub: [bad2212/transformer-model](#) · Hugging Face: [Badalt/fr-en-transformer-scratch](#) · W&B run: [fr-en-transformer / run 5ks4crpz](#) · W&B report: [training and generalization report](#)

1. Summary

A 39.7M-parameter encoder-decoder Transformer (6 encoder / 3 decoder layers, pre-LN, RoPE, tied embeddings, 16k joint SentencePiece) trained from scratch on 923,274 filtered opus-100 pairs for 3.6 h on one Colab T4. Dev OVERALL **39.94**: BLEU 29.8 on *seen*, 34.7 on *long*, 21.9 on *unseen_domain* (literature). **Length is not the failure mode at these lengths; domain is.** The 60-sentence dev slices are too small to prove the gap (its 95% CI includes zero), but 1,000-sentence eval-only samples show it clearly: 31.6 vs 15.8 BLEU. Rare words explain part of it; most of it is register: even books sentences with no rare word trail in-domain ones by ~14.1 BLEU, and the model's loss on literary references is 0.83 nats/token higher.

2. Architecture decisions and the alternatives rejected

Choice	Why	Rejected alternative (trade-off)
Encoder-decoder [1]	Conditional generation: bidirectional source encoding, cross-attention aligns target to source words	Decoder-only: causal view of the source, sequences 2x longer, capacity spent modelling French
6 enc / 3 dec, d=512, 8 heads, FFN 2048 [1, 2]	Measured on the T4: 6/6 (52.3M) ran at 11.5k target tok/s, 6/3 at 13.6k: ~20% more updates in the budget; a shallow decoder also decodes faster [2]	6/6 base: slightly better per step, 15% slower; Transformer-big: ~4x compute, overfits 1M pairs
Pre-LayerNorm + final LN [3]	Stable gradients with a high peak LR (7e-4), no fragile warmup tuning	Post-LN: marginally better when tuned, diverges easily early
RoPE in self-attention; none in cross-attention [4]	Relative positions; cross-attention aligns by content. <i>Not</i> chosen for extrapolation: 8.5% of training pairs exceed 40 words, so <i>long</i> is in range	Sinusoidal (absolute; kept as ablation config), ALiBi [5] (extrapolation is not what <i>long</i> tests), learned (hard length cap)
Three-way tied embeddings [6]	Joint vocab: one matrix for encoder input, decoder input and output layer; saves 16.4M params, regularizes	Separate vocabularies: no tying, names/numbers harder to copy
SentencePiece unigram, 16k joint, byte fallback [7]; source-side subword sampling $\alpha=0.1$ [8]	Robust segmentation of rare/unseen words; no <unk> ever; target stays deterministic	BPE [9] (needs BPE-dropout for the same effect), 32k (fewer examples per rare piece), characters (4-5x longer sequences)
Beam 5, GNMT length penalty $\alpha=1.8$ [10]; split inputs over 128 tokens at sentence ends	Tuned on dev. chrF ($\beta=2$) and BLEU's brevity penalty both punish short output, so a long-output bias pays	Greedy (-1.4 OVERALL); KV cache omitted on purpose (500 sentences; most bug-prone code)
Own PyTorch implementation	RoPE inside attention, pre-LN, tying; ~170 lines, unit-tested (causality, padding, overfit-and-decode)	<code>torch.nn.Transformer</code> : post-LN, no RoPE hook

3. Data and training

Data. opus-100 en-fr train only. Normalised (NFKC; typographic quotes to ASCII, which matters because books sources use ' while their references use '), then filtered: 32,548 duplicates, 21,617 bad length ratios, 14,770 untranslated copies, 6,602 empty, 1,186 over 1,000 chars, and 3 pairs whose source appears in dev/test: **923,274 of 1,000,000 kept** (no subsampling). opus_books was never used for training, the tokenizer, checkpoint choice or decoding tuning; a 1,000-sentence sample (dev/test sentences removed) is used only post hoc in §5.

Training. Label smoothing 0.1 [1], AdamW (0.9, 0.98), 4k warmup + inverse-sqrt, ~24k tokens/update (8k × 3 accumulation), length-bucketed batches, fp16, clip 1.0, seed 42 everywhere. Stopped at step 9,000 (≈8.3 epochs, 3.6 h of a planned 5.5 h) when the free-Colab GPU quota ran out; dev was still improving (table below), so this model is under-trained. Final weights = average of checkpoints 7k-9k [11]: greedy OVERALL 38.49 vs 38.14 for step 9k alone and 37.66 for averaging 5k-9k (early weights hurt while the model still improves fast).

Step	Val loss	Dev OVERALL (greedy)	Beam	α	OVERALL	BLEU	chrF
2,000	4.11	14.4	1	0.0	38.49	28.45	46.81
4,000	2.65	30.8	4	0.6	39.22	29.66	47.22
6,000	2.28	35.4	4	1.0	39.31	29.80	47.30
8,000	2.14	37.6	4	1.4	39.47	30.00	47.50
			5	0.6	39.12	29.61	46.98
			5	1.0	39.23	29.69	47.17
			5	1.4	39.83	30.23	47.77
			5	1.8	39.94	30.23	47.91
			5	2.2	39.83	30.09	47.85

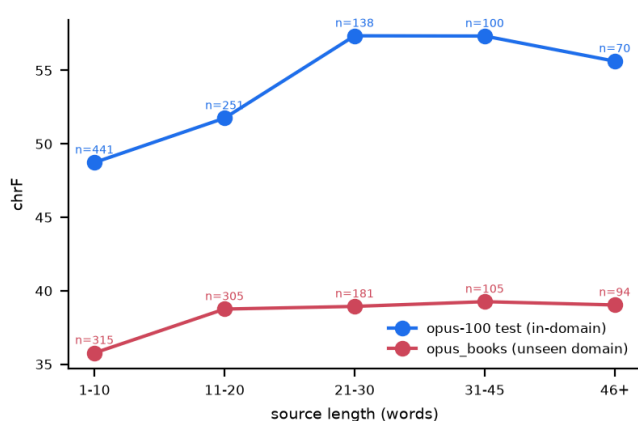
4. Dev results by slice (official scorer, final decoding)

Slice	n	BLEU	95% CI	chrF	95% CI
seen	60	29.8	[21.9, 38.0]	46.4	[40.4, 52.4]
long	30	34.7	[24.9, 43.9]	60.0	[53.9, 66.1]
unseen_domain	60	21.9	[17.1, 26.2]	43.4	[39.3, 48.0]
all	150	30.2		47.9	

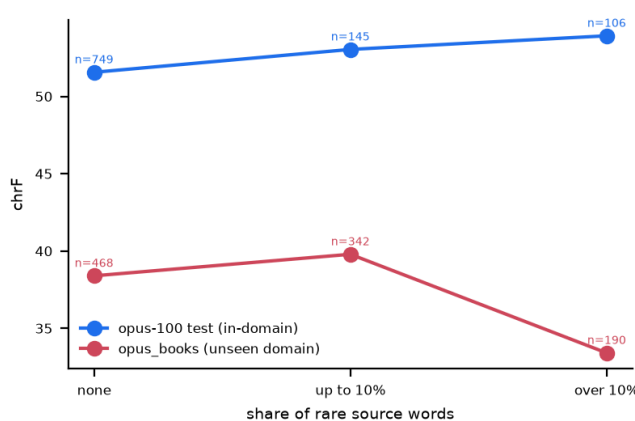
OVERALL = 39.94. CIs: 1,000 bootstrap resamples per slice. The seen – unseen_domain gap is 7.9 BLEU, 95% CI [-1.8, 17.5], and 3.0 chrF, CI [-4.5, 10.6]: **with 60 sentences per slice the dev gap alone is not significant**, which is why §5 measures it on 1,000 sentences per domain. *long* scores *above seen*: long opus-100 sentences are formulaic official text that is well covered in training and within the trained length, while short *seen* sentences are colloquial subtitles with loose references (≈ 4 of 150 dev pairs are misaligned, e.g. dev_00012 → "Outside, Roger.>").

5. Generalization: evidence and hypothesis

Eval-only set (n=1,000 each, post hoc, greedy)	BLEU	chrF	src words	OOV words	rare (<5) words	sent. w/ rare	ref loss/token	hyp/ref length
opus-100 test (in-domain)	31.6	52.0	17.6	1.1%	2.2%	25%	2.05	0.94
opus_books sample (unseen domain)	15.8	37.9	21.4	1.9%	5.1%	53%	2.88	0.90



chrF by source length. The books curve sits below opus-100 in every bucket, so length does not explain the gap.



chrF by share of source words seen fewer than 5 times in training. In-domain, rare words barely matter; in books, sentences with over 10% rare words fall sharply, yet rare-free books sentences already sit far below.

Hypothesis. The gap is register and lexical shift, not length. (1) **Register (largest share):** books sentences with *no* rare word still score 17.8 BLEU vs 31.9 in-domain, and the model's loss on the correct reference is 2.88 vs 2.05 nats/token (perplexity 17.9 vs 7.8): after training on subtitles and official text, 19th-century narrative English is improbable to it. (2) **Vocabulary:** books sources have 1.8x the never-seen-word rate and 2.4x the rare-word rate; the 190 books sentences with over 10% rare words drop to 9.3 BLEU. Names and literary words become frequent words ("Le grand Meaulnes" → "The Great Meals", *cils* (eyelashes) → "cloaks"). (3) **Freer references and shorter outputs:** hypothesis/reference length is 0.90 on books vs 0.94, so the brevity penalty bites more. Typography is ruled out by normalisation. On the 23 eval-only sentences over 128 subword tokens, splitting gives chrF 41.2 vs 39.5 without it (BLEU 19.6 vs 19.0).

6. Where it fails

- **Rare words and names:** subword pieces recombine into a frequent English word ("roulotte" → "roller", "grognaît" → "grew up").
- **Literary word senses:** the common sense wins over the literary one (*fers*, forceps → "irons"). Past tenses are mostly right (*répondit* → "answered", *comprit* → "understood"), with occasional slips (*jeta* → "is throwing").
- **Repetition loops are rare:** 0.5% of in-domain and 0.4% of books outputs repeat a 3-gram three times or more, mostly on long technical or literary inputs.
- **Noisy references** cap scores: misaligned opus-100 pairs, books references with stray quotes and chapter headings.

7. Challenges and how they were resolved

Problem	Resolution
Free Colab disconnects, then the GPU quota ran out at 3.6 h	Atomic full-state checkpoints, deterministic per-epoch batch order (resume skips consumed batches), wall-clock budget summed across sessions, same W&B run; resumed twice. Kill-and-resume was tested before the real run and caught a bug (RNG state moved to GPU by <code>map_location</code>)
T4 slower than planned (11.5k tok/s)	Switched 6/6 → 6/3 by a threshold set before measuring
Noisy data; curly vs straight quotes (one ' costs ~20 chrF on a sentence)	Filtering (7.7% removed) and one normaliser applied to inputs and outputs
Decoding stalls: one 300-word sentence kept 60 × 4 beams decoding to the length cap	Length-sorted batches capped at 2,048 source tokens (outputs unchanged)

Tokenizer not reproducible across machines (12 of 16k pieces differ, 2 vs 8 threads)	Thread count pinned; the trained tokenizer ships with the model
Beam-search edge cases; abbreviation splitting ("M. Dupont")	Unit tests: greedy = beam(1), overfit-and-recover; abbreviation-aware splitter

8. With more time or compute

- Train to convergence (dev still rising at 9k steps) with a cosine tail; several seeds for variance.
- Back-translate monolingual English literature (not opus_books) [12]; domain tags; a copy mechanism for names.
- Run the prepared RoPE vs sinusoidal and 16k vs 32k ablations at matched steps; language-ID and dual cross-entropy data filtering [13].
- KV cache + MBR decoding with chrF as utility; report sacreBLEU alongside the challenge scorer.

References

[1] Vaswani et al. 2017, Attention Is All You Need.	[7] Kudo & Richardson 2018, SentencePiece.
[2] Kasai et al. 2021, Deep Encoder, Shallow Decoder.	[8] Kudo 2018, Subword Regularization.
[3] Xiong et al. 2020, On Layer Normalization in the Transformer Architecture.	[9] Sennrich et al. 2016, Neural MT of Rare Words with Subword Units.
[4] Su et al. 2021, RoFormer: Rotary Position Embedding.	[10] Wu et al. 2016, Google's Neural Machine Translation System.
[5] Press et al. 2022, Train Short, Test Long (ALiBi).	[11] Popel & Bojar 2018, Training Tips for the Transformer Model.
[6] Press & Wolf 2017, Using the Output Embedding to Improve Language Models.	[12] Sennrich et al. 2016, Improving NMT Models with Monolingual Data.
	[13] Junczys-Dowmunt 2018, Dual Conditional Cross-Entropy Filtering.